

An Image Processing Pipeline for Automated Detection of Tomato Leaf Diseases in Greenhouse
Environments

by

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Abstract

Human population growth continues to strain the global food supply, and agricultural production needs to increase by 70% by 2050 to meet this demand. As a result, the development of cultivars that can produce the most yield and are resistant to various adverse abiotic and biotic stressors has become increasingly important. Plant phenotyping methods are employed to investigate gene-environment interactions in the development of such cultivars, and over the past decade, automated image-processing and analysis techniques in plant phenotyping have assisted in these investigations. Despite that, automated image-based tasks can become costly. Consequently, there is a need for the development of low-cost, automated image-based plant phenotyping systems in greenhouse environments. This study develops an open-source image-processing-based plant phenotyping component, which helps to detect various factors of plant health, stress and growth to support low-cost automated greenhouses.

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Introduction

The rapid growth of the human population has created a strain on international food security. Consequently, crop production must increase by 70% by 2050 to meet this demand [1]. Over the past decade, methods in image-processing-based plant phenotyping have been employed to investigate plant gene-environment interactions in greenhouse environments to develop cultivars that exhibit optimal yield and resistance to abiotic and biotic stressors. Despite these advances, current solutions are non-generalizable to a broad range of plant species and can be costly [2, 3, 4]. Thus, there is a critical need for affordable, low-cost plant phenotyping image processing tools in automated greenhouses. This study develops an open-source image-processing-based plant phenotyping component, which helps to detect various factors of plant health, stress and growth to support low-cost automated greenhouses. The low-cost imaging component developed utilises low-complexity and efficient image segmentation techniques suitable for resource-constrained hardware, leveraging open-source software and publicly available labelled plant image datasets. Additionally, our component is extensible and flexible, making it more accessible and generalisable to other plant environment contexts. Following this introduction, we review the literature on image processing in greenhouse environments, paying particular emphasis to plant phenotyping and its applications with image processing in automated, low-cost greenhouse environments. Thereafter, we discuss our methodology for developing the low-cost image processing tool, detailing our automated image processing and analysis pipelines. Lastly, we discuss the effectiveness and cost-effectiveness of the developed component using publicly available plant image datasets.

Literature Review

2.1 Introduction

This literature review provides a foundational understanding of image processing in plant phenotyping and directions for developing or investigating solutions for low-cost image-processing phenotyping tools in automated greenhouses. The focus is on the existing image processing solutions applied in low-cost automated greenhouse environments. We begin by examining the fundamental concepts of plant phenotyping and the various traits that are amenable to image processing for predicting plant health, growth, and stress. Specifically, we note that leaf phenotypic traits are a robust proxy indicator of physiological and metabolic factors affecting plant health [5].

Thereafter, aspects of image processing in plant phenotyping are discussed. Notably, image processing and acquisition, as well as various image sensor technologies and segmentation methods, are reviewed in the context of plant phenotyping. Hyperspectral and RGB imaging are the most common acquisition methods, and segmentation techniques critically influence the overall accuracy, precision, and quality of phenotypical trait analysis [6, 7].

Lastly, the status, challenges, and potential innovations of low-cost automated image processing in greenhouse environments are discussed. In this vein, the literature indicates a lack of affordable and generalizable image processing tools that can analyse and process large amounts of image data on inexpensive, resource-constrained hardware and sensors [8, 4].

2.2 Plant Phenotyping

Abiotic and biotic stressors limit cultivar biomass and yield [9], whilst human population growth strains the global food supply. Plant breeding, the process of inducing desirable, inheritable traits in plants through controlled human action [10], is now crucial in addressing the challenge of providing an adequate global

food supply. Plant breeders require effective techniques to produce germplasm for cultivars that maximise yield and biomass under various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have proven to be an effective tool for plant breeding [11].

2.2.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [12], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [13]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [14, 15, 3].

2.2.2 Plant Phenotyping As Method For Improving Plant Traits

Phenotypical traits serve as proxies in determining whether specific genotypes have favourable traits in particular environmental conditions. Quantitative data produced by plant phenotyping is most helpful in analyzing plant-environment interactions in plant breeding [12] for the development of germplasm with improved plant traits [3]. The data produced by phenotyping is used as a proxy for assessing a plant’s performance in its environment. For example, stressors or disturbances in plants are induced by environmental conditions to which organisms are exposed, and these stressors induce symptoms that can be detected and evaluated using phenotyping. Hence, phenotypical traits of a plant have a complex dynamic feedback loop with their environment [16]. Many phenotypical traits are observable or quantifiable by image processing techniques.

2.2.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

2.2.3.1 Simple Taxonomy of Phenotypical Traits

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [17]. Holistic properties consider the plant as a whole, whereas component-based features examine individual parts of the organism, such as its fruit, flower, leaves, stem, etc.

2.2.3.2 Leaf-Derived Phenotypical Traits: Robust Proxy Indicators of Plant Metabolic and Physiological Status

Leaf-derived traits are among the most used indicators of plant status because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth, and stress [5]. Traits of leaves

can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [18]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [19, 5], and leaf thickness for measuring abiotic stressors like water scarcity [20]. Additionally, physiological traits of leaves exist that indicate various factors of plant status, such as stomatal conductance for detecting biotic stressors like pathogens [21] and leaf water content for drought stress [22]. Considering the above, leaf-derived phenotypical traits are robust proxies of various metabolic and physiological plant parameters.

2.2.3.3 Alternative Phenotypical Traits

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [5]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [23], stems [24], seeds [25], and fruit [26].

2.3 Image-Based Plant Phenotyping

2.3.1 Image-Based Plant Phenotyping Versus Traditional Methods

Image-based plant phenotyping methods have several advantages over traditional methods. Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant’s ontogenesis [2, 5, 27]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. These image processing solutions reduce the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [17]. Despite these advantages, several aspects of image processing, which determine the overall cost, efficiency, and applicability of image processing in plant phenotyping contexts, are important. The most important of these aspects are image acquisition and processing tasks.

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [26], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [28]. For image-based plant

phenotyping, this means that image acquisition and processing techniques are vital.

2.3.2 Image Acquisition Techniques

2.3.2.1 The importance of Image Acquisition in Plant Phenotyping

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired, and the choice of imaging technique for data acquisition affects the types of phenotypical parameters that can be measured. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue, which are not visible to the naked eye [3]. Therefore, the appropriate selection of image acquisition techniques is crucial for analysing different types of plant traits.

2.3.2.2 Common Image Sensor Techniques In Plant Phenotyping

Commonly used image acquisition techniques include visible, fluorescence, thermal, tomographic, and hyperspectral imaging [29, 3]. The most popular techniques are visible and hyperspectral imaging due to their affordability, robustness and flexibility in measuring a wide range of phenotypic parameters. Visible imaging is an affordable option that utilises high-resolution RGB cameras. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [30, 31]. Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, allowing for detailed measurements of various phenotypic parameters. Notable applications of hyperspectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [32]. The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [3, 21], fluorescence imaging in plant pathogenic symptom monitoring [33, 34], and tomography in estimating plant organ water content [35].

2.3.3 Image Processing Techniques

2.3.3.1 The importance of Image Segmentation in Plant Phenotyping

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes, such as *Arabidopsis* [3, 36]. However, segmentation is a key image-processing task common to many image-based plant

phenotyping solutions. Moreover, the validity and precision of the derived phenotypical measurements and analysis depend on the image segmentation method selected [6, 7], because the accuracy of more complex tasks, such as object detection, 3D reconstruction, and classification, depends on segmentation accuracy. Therefore, image segmentation methods are integral to image processing in plant phenotyping.

2.3.3.2 Commonly Used Image Segmentation Techniques

In plant phenotyping, selecting the appropriate segmentation technique is crucial when considering the type of phenotypic parameter to be measured. Segmentation methods are typically classifiable as traditional or learning-based [6]. Traditional segmentation methods typically target holistic phenotypical traits like plant height and include frame differencing [37], colour-based [38], edge detection [39], spectral band difference-based [40], shape modelling-based [41], and graph-based segmentation [42]. In contrast to traditional techniques, learning-based methods, which utilise machine learning or deep learning [43, 44], are more suitable for targeting component-based traits of specific organs or parts of a plant [6].

2.4 Low-cost Automated Image-Based Plant Phenotyping In Greenhouses

Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [45], automated conveyor belts, and benchtop configurations [46, 47], as well as advanced image sensors in high-throughput phenotyping systems [48]. However, these solutions are not accessible to all plant growers because they tend to be costly [49, 50, 4]. Moreover, many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [51]. Furthermore, existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [2, 3]. Therefore, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing systems for plant phenotyping applications.

2.4.1 Open-source Software, Systems and Affordable Hardware As Cost Reducers In Image-Based Plant Phenotyping

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [52, 4] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and

hyperspectral sensors to monitor phenotypic traits in greenhouse environments [50]. Additionally, there has been an increase in open-source image-processing plant phenotyping systems like Phenobox [51]. Despite these advances, several challenges remain in low-cost plant phenotyping systems.

2.4.2 Challenges In Automated Image-Based Plant Phenotyping

2.4.2.1 Large Image Sensor Data Throughput Necessitates Robust Image Processing Tools

Modern image sensors in image-based phenotyping generate massive amounts of data [8], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [4]. Therefore, effective choices in image sensors are crucial for developing low-cost, automated image-based phenotyping systems [5].

2.4.2.2 The Effects of Image Sensor Choice On Phenotypical Analysis And Processing

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [3]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants [53]. Consequently, in low-cost settings, hyperspectral imaging of component-based phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [53]. Additionally, low-cost hardware has constrained bandwidth, storage, and compute power. Therefore, remote cloud image processing and analysis are required [54], which can be expensive [4].

2.4.2.3 Solutions, and Limitations For Low-cost Large Data Throughput Image Processing Systems

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [4], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [50], and image compression techniques [55]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [6]. RGB sensors are affordable and generate less raw

data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [3]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [55]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [7].

2.4.3 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Numerous opportunities exist to improve the overall affordability and generalisability of plant phenotyping in automated greenhouse environments. Low-cost plant phenotyping is not a nascent field of inquiry. However, many tracts of innovation have not been fully exploited. For instance, learning-based imaging techniques for analysis and processing can reduce analysis and segmentation costs but struggle when executed on resource-constrained hardware [28]. Therefore, there is a need for the development of efficient and optimised deep-learning or machine-learning models in resource-constrained environments. Similarly, many existing phenotyping systems are not generalizable to other plant species [3, 36] and rely on proprietary software or hardware [51]. Hence, there are opportunities to develop modifiable and flexible image-processing pipelines that are amenable to a broader variety of plant phenotyping contexts. Additionally, there are not many mobile applications for plant phenotyping. Mobile devices have advanced image sensors and can locally run image analysis and processing tasks, eliminating the need for cloud computing resources that increase phenotyping costs [7].

2.5 Conclusion

Ensuring an adequate food supply in response to the growing human population is crucial [1]. Image-based plant phenotyping has proven to be an effective tool for this purpose, aiding in the development of cultivars with maximum yield and resistance to various environmental stressors [11]. These methods are non-destructive, more time-efficient, replicable, and affordable than traditional methods [17]. However, image sensors in plant phenotyping generate large amounts of data, which necessitate robust image analysis and processing tools to handle this data effectively [8]. Large plant data throughput makes image-based plant phenotyping in automated greenhouse environments cost-prohibitive, as affordable off-the-shelf hardware and sensors often lack adequate local resources to store or process large amounts of data [53]. Consequently, remote cloud computing solutions handle image processing and analysis tasks [54], which can further increase operational costs in greenhouse management [4]. Image-data compression techniques, low-complexity image analysis and processing techniques, and low-cost off-the-shelf hardware are effective in reducing the

cost of image-based plant phenotyping by reducing the computational requirements and data throughput [55, 50, 4]. However, these solutions come with marked trade-offs in terms of accuracy, precision, depth, and breadth of plant trait analysis. Moreover, existing solutions are not generalizable to other environments and plant species [2, 3]. In short, there is a need for low-cost, open-source, flexible, modifiable, and robust plant phenotyping image processing and analysis tools that are effective and efficient on resource-constrained hardware [4]. Future research should develop low-cost image processing tools that are precise, accurate, and adaptable to various plant monitoring contexts or parameters. In doing so, automated image-based techniques for plant monitoring will become more accessible in low-cost greenhouse environments.

Materials and Methods

3.1 Dataset

The PlantVillage dataset is used in our study. PlantVillage is a collection of 54309 annotated diseased and healthy leaf images spanning several plant species, including tomato [56]. The size of the subset of tomato leaf images is 18162. Several images of each tomato leaf were captured outside under natural light in various lighting conditions and orientations, against a black or grey background, using a standard digital camera [56, 57]. The images are captured in the RGB colour space and have dimensions 256x256. We have selected PlantVillage as our primary dataset because it is widely used in image processing research for plant disease detection [58, 59, 60, 61, 62, 63]. Using this dataset will allow future researchers to compare our proposed method to previous studies.

3.2 Image processing pipeline

Many studies in image-based plant disease detection typically have predetermined phases in their image processing pipeline. Namely, preprocessing, segmentation, and feature extraction [60, 64, 65, 66, 67]. We follow suit and include each of those phases in our image processing pipeline.

3.2.1 Preprocessing Phase

Preprocessing involves tasks such as noise removal, image enhancement, and colour space conversion. In image acquisition, noise from the imaging sensor, the environment, or another source can compromise the accuracy of downstream image processing and classification phases. For example, we choose a global thresholding segmentation method, which we discuss later, and these methods often perform poorly in the presence of noise or uneven illumination [68]. Nevertheless, preprocessing steps such as histogram equalisation and median filtering effectively mitigate noise and illumination issues [68, 69, 70]. Thus, preprocessing is a crucial phase of image processing pipelines, which enhances the overall accuracy of segmentation and image classification [57, 71]. Therefore, we apply

several preprocessing steps to enhance the overall effectiveness and accuracy of our segmentation and classification phases.

We aim to compute texture features of the tomato leaf in each image using the grey-level co-occurrence matrix [72], which requires a grey-level image to calculate. Therefore, the first preprocessing step we perform is the conversion from the leaf image’s original RGB colour space to the grayscale colour space.

Once we obtain the grayscale image of the tomato leaf, we apply histogram equalisation. This preprocessing method is used in the literature to add more contrast between the background and foreground of an image [69]. Image contrast enhancement is achieved by uniformly distributing the grey-level intensity values of the image [68], which makes histogram equalisation particularly useful for dealing with various illumination conditions [57].

Lastly, we apply median filtering to remove common salt-and-pepper noise from the image. Median filtering is an effective image enhancement preprocessing step that has been successfully used in plant leaf disease detection [38, 67]. Noise manifests as sharp transitions in image intensity seen in the grayscale image histogram [68], therefore noise can negatively affect the accuracy of global threshold segmentation methods, which we use in this study; therefore, to enhance segmentation accuracy and robustness, we apply median filtering to denoise the image.

3.2.2 Segmentation Phase

- We use a global threshold segmentation method since it is easy to implement, computationally fast, and has been widely used successfully in many image processing segmentation pipelines [68, 73]
- Use this resource to find brief sentence describing otsu method [65]
- Otsu method is a widely applied threshold segmentation method in plant phenotyping. Its primary advantages include simple calculation and high efficiency making it very attractive for low-cost image processing pipelines. It is however sensitive to noise, and has low robustness to varying light conditions where bimodal histogram distributions are not possible. However empirically it has been successfully in the phenotyping of many different plant organs across different species [58] (you can include other sources from this article look at the table summarizing the 2d segmentation methods and where otsu is used). It is a global threshold and therefore has simple implementation. and stable performance and is widely used in plant image segmentation [74]. The otsu method selects best threshold on on maximum interclass variance. Suitable for bimodal distributions, but where unimodal histogram or non-bimodal performance is poor. [74]. We apply preprocessing steps which attempt to normalize the

histogram and and mitigate lumincance issues to increase the effectiveness of our segmentation.

- We apply otsu segmentation thresholding method
- due to varying lighting conditions across all images we use adaptive thresholding

3.2.3 Feature Extraction Phase

- Texture features have proven to be espeically useful for plant pathogen detection [75, 76]
- Textture features are emperically best predictors of tomotoe leaf disease detection [77] since occurenc of plat disaes cause specific changes in texture of the leaft and therefore textetd based features can be used to train a classifier [78]
- The gray level co-occurence methodology is a statisitcal way to descibe the shape by smapling the way gray levels ocuren in relation to other gray levels. And have been applied successfully to detect and clasify plant pathogens [75, 77].
- Caculate gray level co-occurence matrix. Concerned with the spatial statistical distrubiton of gray tones. Based on assumption that the texture information in the overall average spatial relationship the gray tones in the image have with one another. [72]. Following a similar approach to [77] we compute the common energy, contrast, sum of squares, correlation, entropy, sum entropy, cluster shade, and cluster prominence, and homogeneity metrics from the gray level occurrence matrix. These are defined as well in the original paper introducing GLCM and mockhtar.

3.3 Classification Mechanism

- SVM is a machine supervised learning method which has been shown to be an effective classifer of leaf diseases in an image processing pipeline [77].
- We use machine learning Support Vector Machine binary classifier to identify whether or not a leaf is diseased or not. This has shown to be particularly effective for leaf disease detection in the literature [57, 77, 79, 63].
- We pass feature vector calculated from gray level co-occurence matrix
- This feature vector is fed into SVM classifier which determines whether or not original image leaf has disease or not

3.4 Analysis

- Calculate metrics to show reliability of binary SVM classifier.
 - F measure (common metric for binary classifiers) [80, 81]
 - Accuracy (common metric for binary classifiers) [80, 81]
 - Precesion (common metric for binary classifiers) [80, 81]
 - true positive rate (common metric for binary classifiers) [80, 81]
 - true negative rate (common metric for binary classifiers) [80, 81]

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